Webtoon Translator

A browser extension that translates Korean webtoons, and later the web novels they are based on, into English while you read directly on the original site. Project plan: what it does, who it's for, whether it can work, and how to grow it.

The idea in one line: open a webtoon chapter, scroll, and the speech bubbles are already in English, with no screenshots and no uploading one image at a time.

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1. The problem

Many Korean webtoons never get an official English release, or arrive months late. Existing tools like Google Lens work one screenshot at a time, which is painful for a chapter with dozens of panels. Some browser extensions now translate comic images too, but quality, layout and smoothness vary a lot, and that is where there may still be room.

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2. What it does (functionality)

Core features (version 1)

- **Auto-detects comic images** on the page as you scroll.
- **Finds the text** inside speech bubbles and on the art.
- **Reads the Korean** (OCR) and groups it into the right order.
- **Translates with context,** sending the whole chapter's text together so names and tone stay consistent.
- **Erases the original text** and rebuilds the background (inpainting).
- **Typesets the English** neatly inside each bubble.
- **Works directly on the original Korean site** (for example Naver Webtoon) as you read, with no screenshots or uploads.

- **Works ahead of you,** translating the next few panels before you reach them, and remembers results so revisits are instant.

Nice-to-have features (later)

- **Web novel mode:** translate the text of web novels that many webtoons are based on (see section 3).
- A glossary per series (character names, places, slang) that you can edit.
- Sound-effect (onomatopoeia) handling.
- Hover on a bubble to see the Korean, romanization and word meanings (good for language learners).
- A "report bad translation" button that also improves the glossary.

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3. How it works (simple view)

1. **Extension** spots comic images on the page.
2. Sends them to a **server** that runs the AI steps: detect, read, translate, erase, retype.
3. Server returns the translated image.
4. Extension **swaps it in** on the page, and saves it for next time.

The heavy AI runs on a rented GPU server because browsers can't handle all of it well. Images are processed temporarily and not stored.

Working directly on the original sites

The extension runs on the page you are already reading, so there is nothing to upload. Things to plan for:

- **Image access:** the extension needs permission for each site so it can fetch the comic images. Some sites restrict how images load outside their own pages, so test early whether you can get the raw image on Naver Webtoon and Kakao Webtoon.

- **Per-site handling:** every site is built differently (lazy loading, long strips cut into slices, login or age checks). Start with one site, make it work well, then add others.
- **Terms of service:** check what Naver and Kakao allow for automated processing of their pages before releasing.

Web novel mode

Many webtoons start as web novels on sites like Naver Series, Kakao Page and Munpia, and the novel often goes much further in the story than the comic. Novels are mostly plain text on the page, so this needs no bubble detection, OCR or inpainting: the extension reads the page text, translates it in chunks, and shows English in place. It is lighter and cheaper per page than comics.

- **Main challenge:** keeping names, honorifics and tone consistent over long text, so the per-series glossary matters most here.
- **Check first:** some sites may display text in ways that are hard to read from the page (for example as images). Test each site you care about.

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4. Who it's for (audience)

Group	Why they'd use it	Would they pay?
Webtoon readers	Want chapters with no official English version	Rarely, expect free
Web novel readers	Want the full story, which often goes beyond the comic	Sometimes, and costs less to serve
Language learners	Want the Korean text plus meaning	More likely
Independent creators and small publishers	Want to localize their own work	Most likely, and no rights problem
Translation teams	Want a faster first draft to edit	Possibly, but need high quality

5. Market research

What already exists

Search results show several tools that already claim comic translation. Immersive Translate advertises in-panel translation on 40+ platforms, including Kakao Webtoon, with right-click and batch options. Musely and ImageTranslate.AI offer web-based webtoon translation, and a Windows app advertises auto-scroll full-page translation. These are vendor claims, so they need testing before you believe them.

So is there room?

Possibly, but as a **quality and experience** improvement, not a brand-new idea. Winning would mean visibly better bubble detection, cleaner erasing, nicer typesetting, consistent names, and smoother scrolling.

How to check this properly (about a week, free)

1. **Head-to-head test:** run 3-4 Korean chapters through existing tools. Note what fails: missed text, bad fit, smudged art, wrong names, waiting time.
2. **Read reviews** of those tools on the Chrome Web Store and Play Store. Complaints show what people want.
3. **Browse Reddit and Discord** (webtoon and manhwa communities) for "how do you read raws" threads.
4. **Show a prototype** to 10-15 real readers with before/after on a chapter they like. Ask if they'd switch.
5. **Check platform terms** (Kakao, Naver) about automated processing.

Honest outlook: if the existing tools are close to your quality, treat this as a portfolio project. If they fail visibly on things readers care about, there is a real opening. Any chance estimates here are gut feel, not data.

6. Scope: what you get after building it

- **Career:** a full ML pipeline you can demo (detection, OCR, translation, inpainting, deployment). Strong for ML and CV internships, with real tradeoffs to discuss: latency, cost, evaluation.
- **Research:** possible writeup on comic translation quality, or a Korean webtoon text dataset (few good ones exist).
- **Open source:** publish the tool (not any chapters) for visibility and contributors.
- **Web novels:** a text-only mode on the same extension, covering the source stories behind many webtoons.
- **Extensions of the same core:** Japanese and Chinese, live audio subtitles for videos and dramas, translating menus, signs and memes.

7. How to build it

Weeks	Step	What you do
1-2	Single-image pipeline	Detect text, OCR Korean, translate, inpaint, typeset on one strip. Start with existing models before training any.
3	Evaluation set	Pick 5-10 chapters, record failures per stage. This guides your effort and gives you portfolio numbers.
3-4	Webtoon format	Slice long vertical strips, handle text cut at slice edges, stitch back.
4-5	Chrome extension	Find comic images, send them for translation, swap results in, prefetch ahead, cache.
5-6	Backend	API plus a job queue on a GPU server (a pay-per-use provider to start).
7-8	Web novel mode	Read page text, translate in chunks with the glossary, show English in place. Reuses the extension and translation logic.

Concepts you'll learn

object detection · OCR · layout analysis · machine translation · inpainting · text rendering · browser extensions · APIs and queues · caching · evaluation

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8. Making money

Ways to earn

- **Freemium:** free daily page limit, paid for unlimited use or a better model.
- **Credits:** pay per batch of pages, fair when costs scale with usage.
- **Creators and publishers:** paid plan for localizing their own work. Fewer customers, best chance of real revenue.
- **Learner tier:** vocabulary and romanization features.
- **Open source plus hosted version:** code is free, convenience costs money.
- **Donations or sponsors** if the community likes it.

Watch your costs

Every page costs you GPU time and translation API usage. A 60-page chapter adds up, and a heavy free user can cost more than a paying user brings in. **Measure your real cost per page before setting prices,** and use caching so the same page is never processed twice.

Realistic expectation

Early on, aim to cover your own costs. Meaningful income is possible but uncertain, so treat it as a bonus.

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9. Marketing

1. **Before/after demos:** short videos and image comparisons on a real chapter. Visuals sell this better than words.
2. **Go where readers are:** Reddit and Discord communities for webtoons, manhwa and Korean learning. Share, ask for feedback, don't spam.
3. **Private beta** with 10-20 people first, then fix the top complaints.

4. **Chrome Web Store** listing with clear screenshots and an honest description. Extension stores won't bring many users alone.
5. **Short-form content:** TikTok, Reels or Shorts clips showing a raw page turning English.
6. **Open-source launch** on GitHub, plus posts on developer communities, for a second audience.
7. **Learn from users:** the "report bad translation" button doubles as free research.

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10. Risks and legal notes

- **Copyright:** translating images on the user's own screen is generally defensible, but hosting or sharing translated chapters is infringement. Don't build a library or sharing feature.
- **Site access:** sites may block or restrict how images and text are read outside their own pages. Test on your chosen site before building around it.
- **Platform terms:** Kakao, Naver and extension stores may restrict tools that look like piracy aids. Frame it as a reading aid.
- **Privacy:** process images temporarily and say so publicly.
- **Before charging money:** get proper legal advice. This document is not legal advice.
- **Quality risk:** bad translations or ugly output lose users fast.

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11. Future plans

1. **Quality:** series glossaries, sound effects, better fonts and bubble fitting.
2. **Web novels:** add text translation for Korean web novel sites right after comics work on one site.
3. **More languages:** Japanese manga, then Chinese manhua.
4. **Mobile:** an app or in-app browser, since many people read on phones.
5. **Live audio subtitles** for videos and dramas, reusing the translation and overlay logic.
6. **Creator tools:** batch export and editing for people localizing their own work.

7. **Learning mode:** word breakdowns and saved vocabulary.

12. How to scale it

Stage	Users	What to do
0. Prototype	You and friends	Run on your own machine or one cheap GPU. Prove quality on test chapters.
1. Private beta	10-50	One pay-per-use GPU server, simple queue, caching, error reporting. Track cost per page and failure types.
2. Public launch	Hundreds to low thousands	Store listing, free tier with limits, payments, rate limiting, monitoring. Shared cache so a popular page is translated once for everyone.
3. Growth	Thousands and up	Autoscaling GPU workers, smaller or faster models for easy pages and bigger ones for hard pages, batch processing, cheaper translation routing.
4. Expansion	Larger audience	New languages, mobile app, creator and team plans, partnerships with publishers.

Levers that matter most when scaling

- **Cache aggressively:** the same popular chapter shouldn't be processed twice.
- **Cut cost per page:** smaller models, batching, cheaper GPUs, and routing easy pages to lighter pipelines.
- **Keep latency low:** prefetch ahead of the reader and process in parallel.
- **Measure quality continuously:** keep your test set and rerun it after every model change.
- **Move on a stage only when the previous one works:** good quality first, then reliability, then growth.

13. Suggested next steps

1. Run the head-to-head test on existing tools (a weekend).
2. Check that you can read images and text on your chosen site (Naver Webtoon first).
3. Build the single-image pipeline on one Korean strip.
4. Show before/after to 10-15 readers and note reactions.
5. Decide: portfolio project, free open-source tool, or a paid product.